# Capacity Portfolio Evaluation Under Multi-Dimensional Uncertainty: A Stochastic Simulation Study

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## 1 Introduction

Long-term electricity capacity planning requires selecting a resource mix that balances cost and reliability across plausible future conditions. In Ontario, the IESO projects a 75% increase in provincial electricity demand by 2050, driven by electrification, electric-vehicle supply chains, data centres, and industrial expansion [Independent Electricity System Operator, 2025]. As the generation mix shifts toward variable renewables and battery storage, deterministic planning methods that evaluate a small number of fixed scenarios may not reveal when uncertainty changes which resource mix performs best.

Recent literature emphasizes explicit uncertainty treatment in capacity planning. Roald et al. [2023] review optimization under uncertainty in power systems. Stover et al. [2023] argue that deterministic reserve-margin rules are insufficient for high-renewable systems. Wang et al. [2024] show that temporal correlation between demand, renewables, and storage materially affects capacity credit. Leibowicz et al. [2024] find that capturing operational detail in Monte Carlo adequacy simulation changes which resources are identified as adequate.

A particularly relevant contribution is Park and Baldick [2016], who formulate a multi-year stochastic capacity expansion model and show that uncertainty in demand and wind availability changes optimal build timing and resource mix. The present study adapts this uncertainty-aware logic to a different problem: rather than optimizing expansion decisions, it evaluates a systematically generated set of fixed candidate portfolios using Monte Carlo simulation and formal statistical selection. Park and Baldick solve a multi-stage stochastic program; this project takes candidate mixes as given and asks which performs best under uncertainty, using Common Random Numbers (CRN), the Kim–Nelson (KN) sequential elimination procedure [Kim and Nelson, 2001], and Bonferroni-corrected confidence intervals.

Candidate portfolios are generated by discretizing the feasible allocation simplex over firm generation, renewable generation, and battery storage at 10% increments subject to a fixed total capacity budget. The simulation model is a stylized one-zone system, not calibrated to a specific

jurisdiction, with input distributions calibrated from Ontario empirical data (IESO demand records and the CODERS dataset [Energy Modelling Hub, 2025]).

The remainder of this report is organized as follows. Section 2 defines the problem and performance measure. Section 3 describes the simulation model. Section 4 presents the experimental design. Section 5 reports the results. Section 6 discusses limitations and future work.

# 2 Problem Description

## 2.1 Portfolio Generation

The system comprises three technology classes: firm generation (nuclear, flexible firm, mid-merit gas, gas peakers), variable renewables (wind, solar), and battery storage. Total installed capacity is fixed at $C_{\text{total}} = 12{,}000$ MW, derived from 8,000 MW peak demand and a 50% reserve margin. Each portfolio $P_i$ is defined by $(f_i^{\text{firm}}, f_i^{\text{renew}}, f_i^{\text{stor}})$ with $f_i^{\text{firm}} + f_i^{\text{renew}} + f_i^{\text{stor}} = 1$.

The feasible simplex is discretized at 10% increments subject to $f^{\text{firm}} \in [0.20, 0.70]$, $f^{\text{renew}} \in [0.10, 0.60]$, $f^{\text{stor}} \in [0.10, 0.30]$, yielding $k = 16$ portfolios (Appendix B). This systematic approach avoids selection bias and ensures uniform coverage of the design space.

## 2.2 Performance Measure

For portfolio $P_i$ in replication $j$:

$$Y_{ij} = F_i + C_{ij} + \lambda U_{ij}, \tag{1}$$

where $F_i$ is the annualized fixed cost (deterministic), $C_{ij}$ is simulated annual operating cost, $U_{ij}$ is unserved energy (MWh), and $\lambda = \$10{,}000/\text{MWh}$ is the value of lost load. Operating cost is computed from hourly dispatch:

$$C_{ij} = \sum_{t=1}^{8760} \sum_{g \in \mathcal{G}} c_g \, x_{g,i,j,t}, \tag{2}$$

where $c_g$ is marginal cost and $x_{g,i,j,t}$ is dispatched output in hour $t$. Both $C_{ij}$ and $U_{ij}$ emerge from the simulation described in Section 3; they are not sampled directly. The objective is $i^* = \arg\min_i \theta_i$ where $\theta_i = \mathbb{E}[Y_i]$.

## 2.3 Research Questions

1. Which portfolio minimizes expected total cost under uncertainty, and does this differ from the deterministic recommendation?
2. Which uncertainty factors materially affect portfolio ranking?

# 3 Model Description

## 3.1 System Architecture

Each portfolio allocates capacity across four firm sub-types, two renewable technologies, and battery storage. Firm generators are dispatched in merit order (Table 1); renewables have zero marginal cost; storage follows a rule-based heuristic. Full parameter values are in Appendix A.

Table 1: Firm generator merit-order stack.

| Sub-type | Share | MC ($/MWh) | FOR | Capital ($/kW-yr) |
|---|---|---|---|---|
| Nuclear | 20% | 22 | 5% | 400 |
| Flexible firm | 35% | 35 | 6% | 100 |
| Mid-merit gas | 30% | 50 | 5% | 70 |
| Gas peaker | 15% | 80 | 8% | 70 |

Nuclear marginal cost of $22/MWh reflects the variable operating cost of refurbished Ontario nuclear units (fuel plus variable O&M), not the all-in levelized cost of electricity.

## 3.2 Hourly Dispatch Algorithm

Each replication simulates 8,760 hours. For each hour $t$:

1. **Demand.** $D(t) = D_{\text{base}}(t)(1 + \sigma_d \varepsilon(t))$, $\varepsilon(t) \sim N(0, 1)$, $\sigma_d = 0.09$.
2. **Renewables.** Wind and solar CFs from NORTA AR(1) Beta distributions (Section 3.3). $R(t) = \text{MW}_w \cdot \text{CF}_w(t) + \text{MW}_s \cdot \text{CF}_s(t)$.
3. **Residual demand.** $D_{\text{res}}(t) = \max(0, D(t) - R(t))$.
4. **Storage.** Discharge when residual demand exceeds the 70th percentile of base demand; charge during off-peak (below 30th percentile). Stylized rules, not co-optimized.
5. **Firm dispatch.** Merit-order dispatch with $x_g(t) = \min(\text{cap}_g \cdot a_g(t), D_{\text{rem}}(t))$, where $a_g(t)$ is outage-adjusted availability.
6. **Unserved energy.** $u(t) = \max(0, D_{\text{rem}}(t))$.

Annual outputs: $C_{ij} = \sum_t \sum_g c_g x_{g,t}$, $U_{ij} = \sum_t u(t)$, and $Y_{ij} = F_i + C_{ij} + \lambda U_{ij}$.

## 3.3 Input Models and Empirical Calibration

Input distributions are calibrated from IESO hourly Ontario demand [Independent Electricity System Operator, 2025] and CODERS hourly capacity factors for 2,984 Ontario generation sites [Energy Modelling Hub, 2025].

**Demand.** The base profile $D_{\text{base}}(t)$ is fitted via OLS to 2024 IESO data using two seasonal harmonics (annual and semi-annual, capturing Ontario's winter/summer demand peaks) plus a diurnal harmonic, achieving $R^2 = 0.59$. The noise parameter $\sigma_d = 0.09$ is the standard deviation of relative residuals $(D_{\text{obs}} - D_{\text{base}})/D_{\text{base}}$.

**Wind.** Hourly aggregate CFs are fitted to Beta$(6.73, 15.69)$ via MLE (KS $= 0.021$, mean $= 0.300$). To preserve the strong temporal persistence (empirical lag-1 ACF $\hat{\rho} = 0.995$), CFs are generated via NORTA with AR(1) dependence:

$$Z(t) = \rho Z(t-1) + \sqrt{1-\rho^2}\,\epsilon(t), \quad \text{CF}(t) = F^{-1}_{\text{Beta}}(\Phi(Z(t))). \tag{3}$$

This preserves the Beta marginal while inducing hourly autocorrelation. The distinction from iid sampling is consequential: iid draws overestimate hourly variability while underestimating sustained low-output periods that drive adequacy risk (Appendix E).

**Solar.** $\text{CF}_s(t) = E(t) \cdot K(t)$, where $E(t)$ is a sinusoidal diurnal envelope (hours 7–21) and $K(t) \sim$ Beta$(1.77, 6.91)$ via NORTA AR(1) with $\rho = 0.928$, calibrated from CODERS.

**Forced outages.** Each firm sub-type has 10 units with independent daily Bernoulli$(1 - \text{FOR})$ draws, producing discrete availability steps of 0%, 10%, ..., 100%.

# 4 Experimental Design

## 4.1 Common Random Numbers

All portfolios share identical random inputs within each replication, implemented via NumPy's `SeedSequence` spawning. Three independent streams (demand, outages, renewables) are deterministically generated from a base seed and replication index. CRN induces positive covariance between outputs, reducing $\text{Var}(Y_i - Y_\ell)$. Empirically, variance reduction ranges from 2× to 35× (Appendix, Figure 10).

## 4.2 Kim–Nelson Sequential Elimination

The KN procedure [Kim and Nelson, 2001] selects the portfolio with smallest $\theta_i$, guaranteeing $P(\text{CS}) \geq 1 - \alpha$ when the best system is at least $\delta$ better than the second-best. Parameters: $\alpha = 0.05$, $\delta = \$50\text{M}$, $n_0 = 30$. The procedure initializes with $n_0$ CRN replications from all $k = 16$ portfolios, computes pairwise difference variances $S^2_{i\ell}$ (which automatically capture CRN covariance), then sequentially eliminates portfolios whose running means exceed a competitor's beyond a shrinking threshold. The procedure terminates when one survivor remains.

### 4.3 Deterministic Benchmark Gap

The deterministic recommendation $P_{\text{DET}}$ is obtained by setting all inputs to expected values and selecting the lowest-$Y$ portfolio. $P_{\text{DET}}$ and $P_{\text{KN}}$ are then evaluated for 200 paired CRN replications. The DBG $= \bar{Y}_{\text{DET}} - \bar{Y}_{\text{KN}}$ quantifies the cost of ignoring uncertainty in portfolio selection. This is not the classical VSS (which compares stochastic and deterministic optimizations); since we select from fixed portfolios, the DBG measures the value of uncertainty-aware selection.

### 4.4 Sensitivity Analysis

A $2^3$ factorial tests demand noise ($\sigma_d \in \{0.03, 0.15\}$), forced outage rates (all 1% vs. realistic), and renewable variability (fixed mean vs. stochastic), restricted to the top-4 portfolios with 30 CRN replications per cell.

### 4.5 Extensions

Two additional analyses extend the core evaluation. First, a *stochastic adequacy search* finds, for each top-4 mix, the minimum total capacity such that EUE $\leq$ 10 MWh/year via bisection (50 replications per step). Second, a *UCAP-normalized comparison* re-evaluates six portfolios (spanning 20–70% firm) each sized to equal effective capacity (8,000 MW UCAP) using ELCC values from IESO planning conventions (Appendix D). This tests whether the equal-installed-MW design biases results toward firm-heavy portfolios. Finally, a *post-processing VOLL sensitivity* recomputes $Y = F + C + \lambda U$ over $\lambda \in \{5{,}000, 10{,}000, 15{,}000\}$ using the same replication-level components from the extended CRN run.

## 5 Results

### 5.1 Portfolio Selection

The KN procedure selects P04 (70/20/10) after 77 replications (Figure 1). Fourteen of sixteen portfolios are eliminated at the initial stage ($n_0 = 30$); only P01 (40/50/10) survives to stage 77. Table 2 reports the extended evaluation.

Table 2: Top-4 portfolio evaluation (200 replications).

| Portfolio | Mean $Y$ (\$M) | $F_i$ (\$M) | $\bar{C}$ (\$M) | $\lambda\bar{U}$ (\$M) | RHW |
|---|---|---|---|---|---|
| P04 (70/20/10) | 4,306 | 1,694 | 1,835 | 777 | 0.38% |
| P10 (70/10/20) | 5,060 | 1,732 | 1,874 | 1,454 | 0.40% |
| P03 (60/30/10) | 8,272 | 1,660 | 1,846 | 4,766 | 0.83% |
| P09 (60/20/20) | 10,785 | 1,698 | 1,882 | 7,205 | 0.64% |

The cost decomposition (Figure 2) explains the result. P04 has the highest fixed cost among competitive portfolios but the lowest adequacy penalty. Portfolios with less firm capacity incur

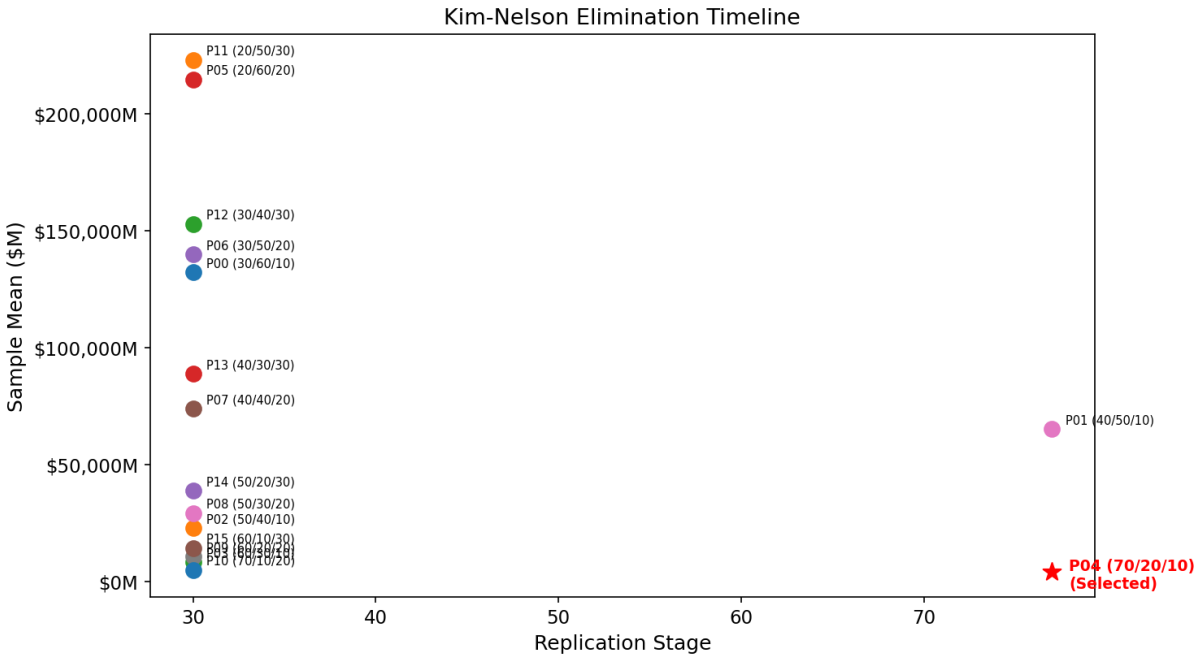


Figure 1: KN elimination timeline. Fourteen portfolios are eliminated at $n_0 = 30$; P01 survives to stage 77 before elimination. P04 (70/20/10) is selected as the best portfolio.

dramatically higher penalties because sustained low-renewable periods, captured by the NORTA AR(1) model, leave insufficient dispatchable generation. All six pairwise differences are significant under Bonferroni correction ($\alpha_{\text{pair}} = 0.0083$); all RHWs are below 1%.

### 5.2 Deterministic Benchmark Gap

Under deterministic assumptions, the best portfolio is P03 (60/30/10) at \$3,503M, which differs from the stochastic recommendation (P04). The DBG is

$$\text{DBG} = \bar{Y}_{\text{P03}} - \bar{Y}_{\text{P04}} = \$3{,}968\text{M}, \quad 95\% \text{ CI: } [\$3{,}905,\ \$4{,}031\text{M}]. \tag{4}$$

A planner using deterministic analysis would select a portfolio costing \$4B more per year under uncertainty (Figure 3). P03 saves \$34M in fixed cost but its lower firm fraction exposes it to adequacy risk invisible to deterministic evaluation.

### 5.3 Sensitivity Analysis

The factorial (Figure 4) identifies demand noise (\$2,883M main effect) and forced outages (\$2,838M) as dominant. Renewable variability contributes only \$117M, reflecting the concentrated Beta(6.73, 15.69) calibrated from fleet-level data. The demand×outage interaction (−\$699M) is substantial. P04 is the best portfolio in all eight configurations, indicating the recommendation is robust to uncertainty assumptions within the tested range.

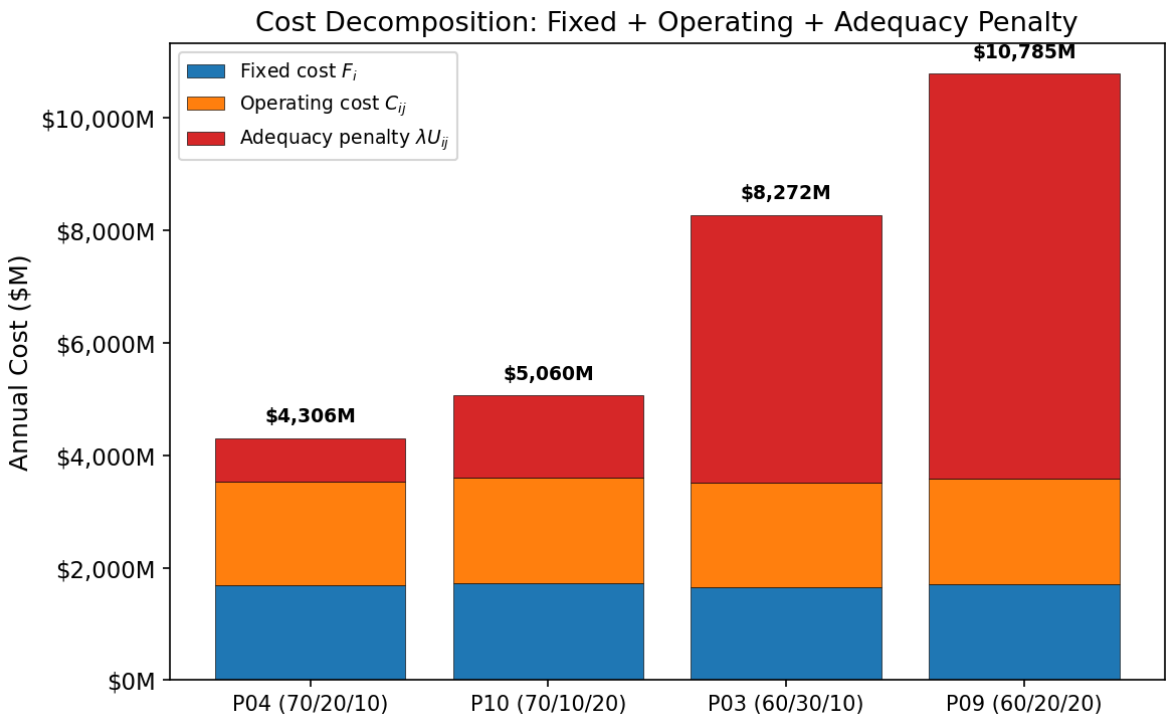


Figure 2: Cost decomposition: fixed cost $F_i$ (blue), operating cost $C_{ij}$ (orange), and adequacy penalty $\lambda U_{ij}$ (red). P04 pays the highest fixed cost but the lowest penalty, yielding the lowest total.

### 5.4 VOLL Sensitivity

To test robustness to the adequacy penalty coefficient, the top-4 portfolios were re-ranked for $\lambda \in \{5{,}000, 10{,}000, 15{,}000\}$ \$/MWh using the same replication-level $(F, C, U)$ outputs from the 200-replication CRN experiment. The selected portfolio remains unchanged (P04) across all tested values. This indicates the recommendation is not an artifact of a single VOLL calibration point, at least within a practically relevant range.

### 5.5 Adequacy Search

When total capacity is a decision variable, P04's mix achieves adequacy (EUE $\approx 0$) at 15.8 GW, while P03 requires 17.5 GW and P09 requires 18.2 GW. The cheapest adequate portfolio is P04 at \$3,763M (Appendix, Figure 12).

### 5.6 UCAP-Normalized Comparison

To test whether equal-installed-MW comparisons bias toward high-firm portfolios, six mixes spanning 20–70% firm are evaluated at equal UCAP (8,000 MW). Even after normalization, P04 remains cheapest (\$9,186M vs. \$10,167M for P03). The adequacy penalty dominates for low-firm mixes despite higher installed capacity, because ELCC is a static metric that does not capture sustained

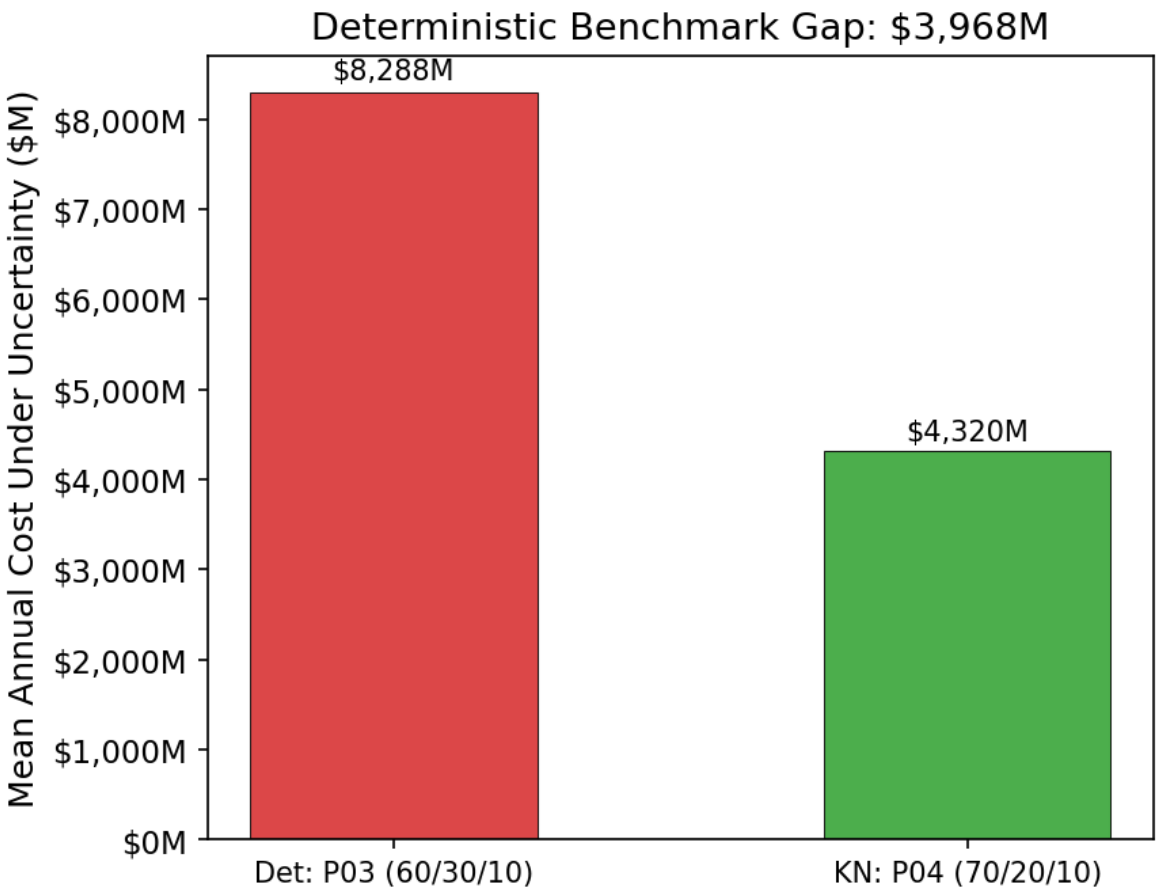


Figure 3: Deterministic Benchmark Gap. Under mean-input assumptions, P03 appears cheaper. Under stochastic evaluation with 200 paired CRN replications, P04 costs $3,968M less per year.

low-renewable events. During multi-hour wind droughts ($\rho = 0.995$), effective renewable UCAP drops to near zero, and additional installed capacity provides no incremental adequacy value (Appendix, Figure 13).

## 6 Conclusion

This study demonstrates a simulation-based framework for comparing capacity portfolios under uncertainty, combining chronological Monte Carlo simulation with empirically calibrated inputs, CRN, and Kim–Nelson ranking and selection.

The central finding is that uncertainty changes the preferred portfolio. Deterministic analysis selects P03 (60/30/10); stochastic analysis selects P04 (70/20/10), which costs $4B less per year when uncertainty is included (DBG = $3,968M, $p < 0.05$). This is robust across all factorial configurations and persists under UCAP normalization.

Three methodological contributions merit emphasis. First, the NORTA AR(1) model preserves both the empirical marginal and temporal autocorrelation ($\rho = 0.995$) of Ontario wind data;

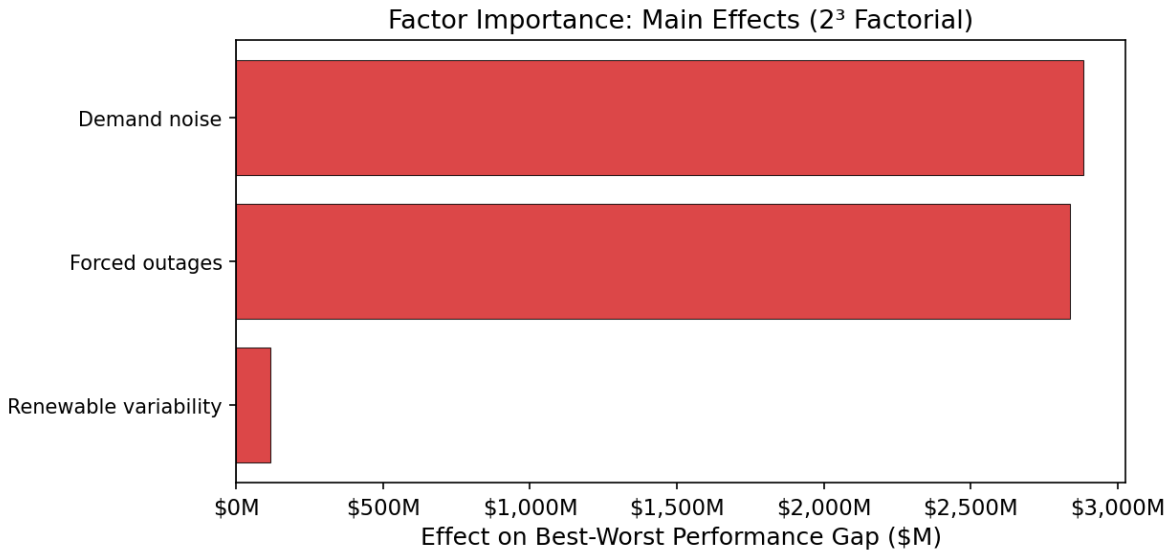


Figure 4: Factorial sensitivity analysis. Demand noise and forced outages dominate; renewable variability has minimal impact. P04 is the best portfolio in all eight configurations.

iid sampling underestimates sustained low-output events. Second, CRN achieves 2–35× variance reduction, substantially accelerating KN elimination. Third, the adequacy search demonstrates stochastic root-finding for minimum reliable capacity, bridging evaluation with expansion planning.

**Limitations.** (1) Demand noise is iid, which may overstate extreme event frequency. (2) Storage dispatch is rule-based, not co-optimized. (3) NORTA AR(1) assumes normal-space $\rho$ equals the target autocorrelation; the error is negligible at $\rho = 0.995$. (4) Discrete outage granularity (10 units per sub-type).

**Future work.** The natural extension is full stochastic capacity expansion, jointly optimizing the portfolio mix and total capacity subject to an adequacy constraint. The adequacy search solves a one-dimensional version; the multi-dimensional formulation connects directly to Park and Baldick [2016].

## References

Energy Modelling Hub. CODERS: Canadian open-source database for energy research and systems-modelling. `https://cme-emh.ca/en/coders/`, 2025. Release October 2025. Hourly provincial generation, demand, and capacity factor data.

Independent Electricity System Operator. 2025 annual planning outlook. Technical report, Independent Electricity System Operator (IESO), April 2025. URL `https://ieso.ca/-/media/Files/IESO/Document-Library/planning-forecasts/apo/2025/2025-Annual-Planning-Outlook.pdf`.

Seong-Hee Kim and Barry L. Nelson. A fully sequential procedure for indifference-zone selection in simulation. *ACM Transactions on Modeling and Computer Simulation*, 11(3):251–273, 2001. doi: 10.1145/502109.502111.

Benjamin D. Leibowicz, Nan Zhang, Juan P. Carvallo, Peter H. Larsen, Thomas Carr, and Sunhee Baik. The importance of capturing power system operational details in resource adequacy assessments. *Electric Power Systems Research*, 228:110057, 2024. doi: 10.1016/j.epsr.2023.110057.

Heejung Park and Ross Baldick. Multi-year stochastic generation capacity expansion planning under environmental energy policy. *Applied Energy*, 183:737–745, 2016. doi: 10.1016/j.apenergy.2016.08.164.

Line A. Roald, David Pozo, Anthony Papavasiliou, Daniel K. Molzahn, Jalal Kazempour, and Antonio J. Conejo. Power systems optimization under uncertainty: A review of methods and applications. *Electric Power Systems Research*, 214:108725, 2023. doi: 10.1016/j.epsr.2022.108725.

O. Stover, P. Karve, and S. Mahadevan. Reliability and risk metrics to assess operational adequacy and flexibility of power grids. *Reliability Engineering & System Safety*, 231:109018, 2023. doi: 10.1016/j.ress.2022.109018.

R. Wang, S. Wang, G. Geng, and Q. Jiang. Multi-time-scale capacity credit assessment of renewable and energy storage considering complex operational time series. *Applied Energy*, 355:122382, 2024. doi: 10.1016/j.apenergy.2023.122382.

# A Consolidated Parameters

Table 3: Simulation parameters.

| Parameter | Value | Basis |
|---|---|---|
| *System* | | |
| Peak demand | 8,000 MW | Stylized |
| Reserve margin | 50% | Accounts for renewable ELCC |
| Total capacity | 12,000 MW | Peak $\times$ 1.5 |
| *Demand* | | |
| Profile | 2-harmonic OLS | IESO 2024 ($R^2 = 0.59$) |
| Noise $\sigma_d$ | 0.09 | Residual std from fit |
| *Renewables* | | |
| Wind CF | Beta(6.73, 15.69) | MLE, CODERS Ontario |
| Wind $\rho$ | 0.9945 | Lag-1 ACF, CODERS |
| Solar cloud | Beta(1.77, 6.91) | MLE, CODERS Ontario |
| Solar $\rho$ | 0.928 | Lag-1 ACF, CODERS |
| Daylight hours | 07:00–21:00 | CODERS (UTC−5) |
| Wind/solar split | 60/40 | Stylized |
| *Storage* | | |
| Duration | 4 hours | Industry standard |
| Round-trip eff. | 85% | NREL ATB |
| Discharge/charge | 70th/30th pctile | Stylized rules |
| *Cost* | | |
| VOLL ($\lambda$) | \$10,000/MWh | Mid-range |
| Nuclear capital | \$400/kW-yr | High overnight, 30-yr |
| Flex firm capital | \$100/kW-yr | Moderate |
| Mid-merit capital | \$70/kW-yr | Combined cycle |
| Peaker capital | \$70/kW-yr | Simple cycle |
| Wind capital | \$130/kW-yr | Onshore |
| Solar capital | \$100/kW-yr | Utility-scale |
| Storage capital | \$150/kW-yr | Battery, 15-yr life |
| *KN procedure* | | |
| $\alpha$ / $\delta$ / $n_0$ | 0.05 / \$50M / 30 | |

# B Portfolio Definitions

Table 4: All 16 candidate portfolios (12,000 MW total).

| ID | Label | Firm | Renew | Storage | $F_i$ ($M) |
|---|---|---|---|---|---|
| P00 | 30/60/10 | 3,600 | 7,200 | 1,200 | 1,557 |
| P01 | 40/50/10 | 4,800 | 6,000 | 1,200 | 1,591 |
| P02 | 50/40/10 | 6,000 | 4,800 | 1,200 | 1,625 |
| P03 | 60/30/10 | 7,200 | 3,600 | 1,200 | 1,660 |
| P04 | 70/20/10 | 8,400 | 2,400 | 1,200 | 1,694 |
| P05 | 20/60/20 | 2,400 | 7,200 | 2,400 | 1,561 |
| P06 | 30/50/20 | 3,600 | 6,000 | 2,400 | 1,595 |
| P07 | 40/40/20 | 4,800 | 4,800 | 2,400 | 1,630 |
| P08 | 50/30/20 | 6,000 | 3,600 | 2,400 | 1,664 |
| P09 | 60/20/20 | 7,200 | 2,400 | 2,400 | 1,698 |
| P10 | 70/10/20 | 8,400 | 1,200 | 2,400 | 1,732 |
| P11 | 20/50/30 | 2,400 | 6,000 | 3,600 | 1,600 |
| P12 | 30/40/30 | 3,600 | 4,800 | 3,600 | 1,634 |
| P13 | 40/30/30 | 4,800 | 3,600 | 3,600 | 1,668 |
| P14 | 50/20/30 | 6,000 | 2,400 | 3,600 | 1,702 |
| P15 | 60/10/30 | 7,200 | 1,200 | 3,600 | 1,736 |

# C Comparison with Park and Baldick

Table 5: Comparison with Park and Baldick [2016].

| Dimension | Park & Baldick | This study |
|---|---|---|
| Objective | Optimize expansion | Evaluate fixed portfolios |
| Method | Multi-stage stoch. prog. | Monte Carlo + KN R&S |
| Demand | Scenario-based growth | Calibrated profile + noise |
| Renewables | Scenario-based wind | NORTA AR(1) Beta CFs |
| Temporal | Representative periods | 8,760-hr chronological |
| Selection | Optimization | Statistical ranking |

Both studies demonstrate that uncertainty can change the preferred resource mix. Park and Baldick show this for expansion timing; this study shows it for portfolio ranking under a fixed-budget constraint using formal statistical selection.

# D ELCC Values

Table 6: ELCC values for UCAP normalization.

| Technology | ELCC | Basis |
|---|---|---|
| Nuclear | 94% | High availability baseload |
| Flexible firm | 92% | Gas/hydro, slight derates |
| Mid-merit gas | 95% | CCGT |
| Gas peaker | 92% | Cycling stress |
| Wind | 18% | IESO capacity auction |
| Solar | 35% | Summer peak contribution |
| Storage (4h) | 50% | Duration-limited |

# E Input Fitting Diagnostics

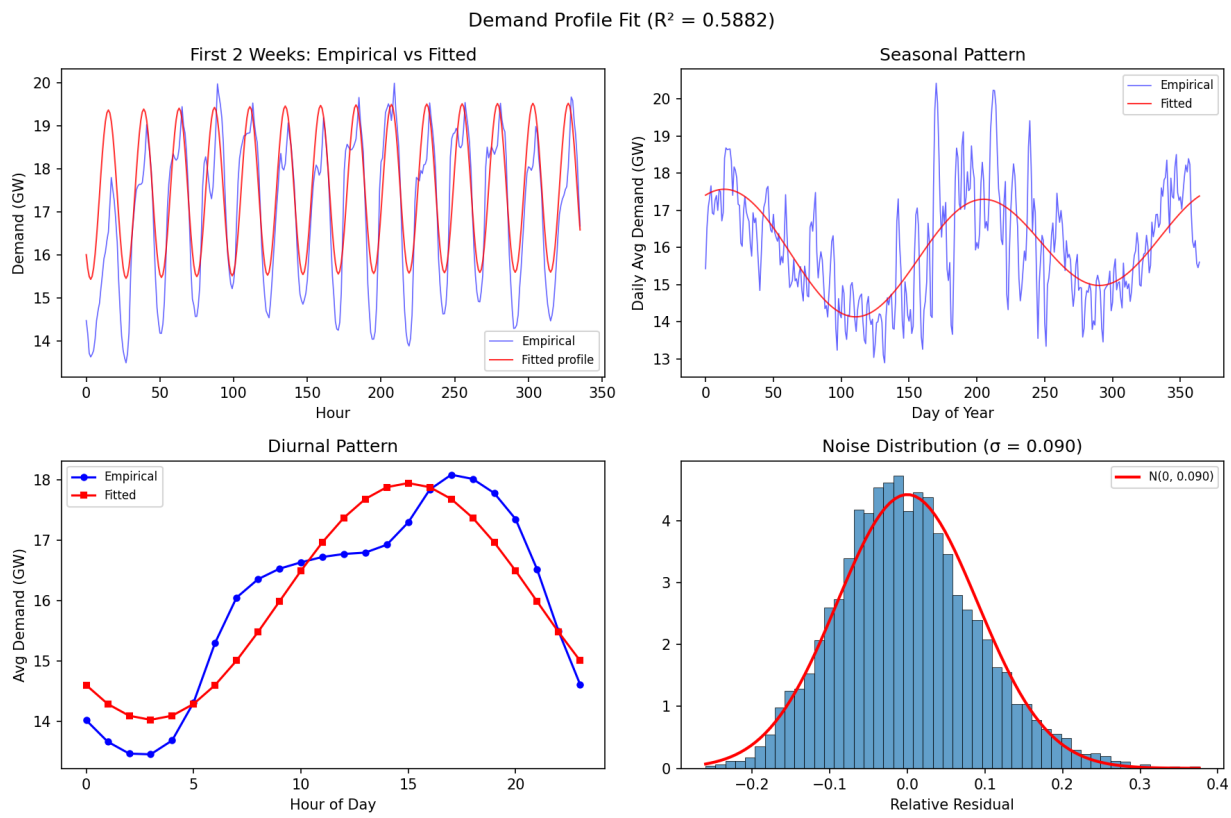


Figure 5: Demand profile OLS fit to IESO 2024 hourly Ontario demand. Left: observed vs. fitted. Right: residual distribution ($\sigma_d = 0.09$).

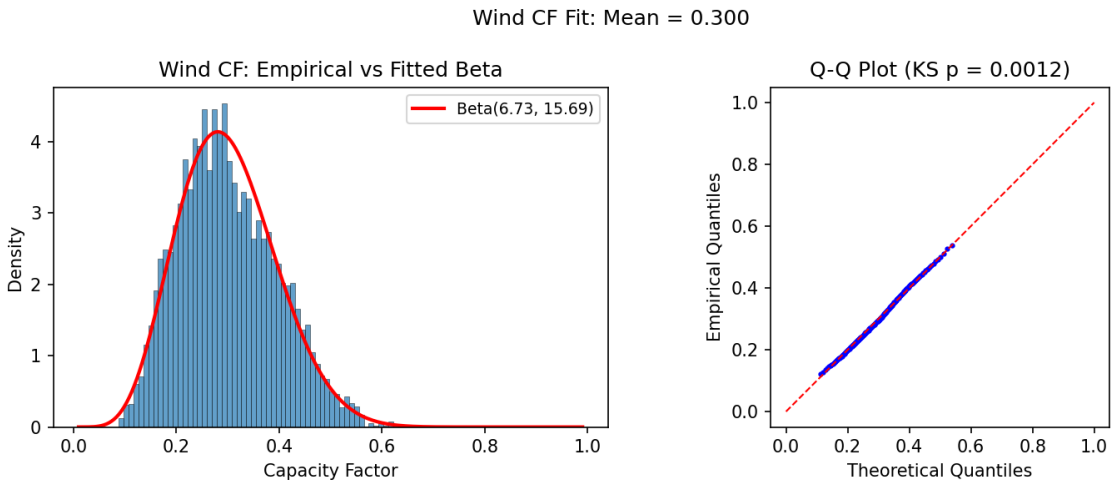


Figure 6: Wind CF Beta(6.73, 15.69) MLE fit to CODERS Ontario. Left: empirical vs. fitted PDF. Right: Q–Q plot (KS = 0.021).

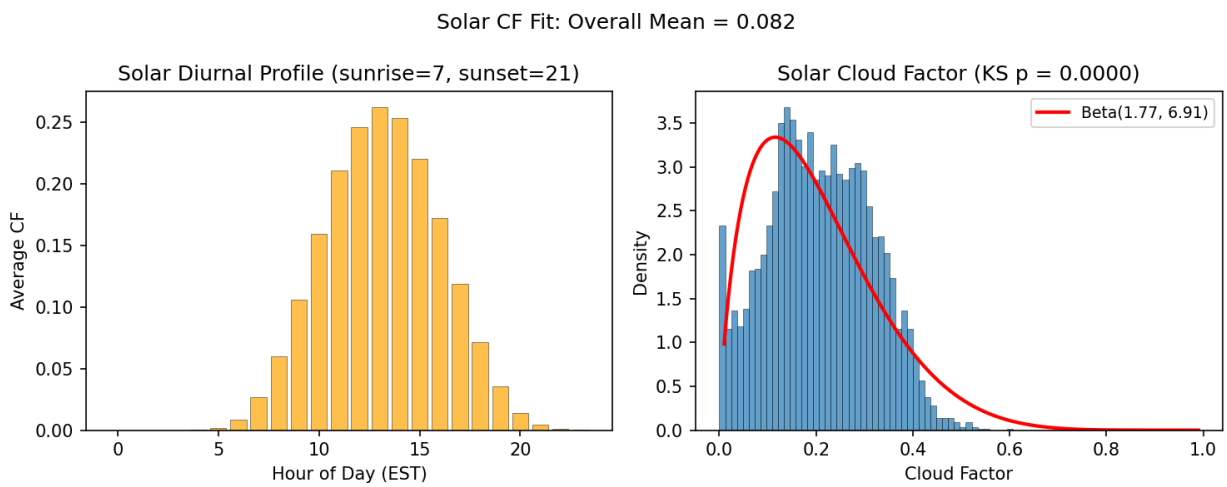


Figure 7: Solar CF: diurnal envelope × Beta(1.77, 6.91), calibrated from CODERS Ontario.

# F Supplementary Results

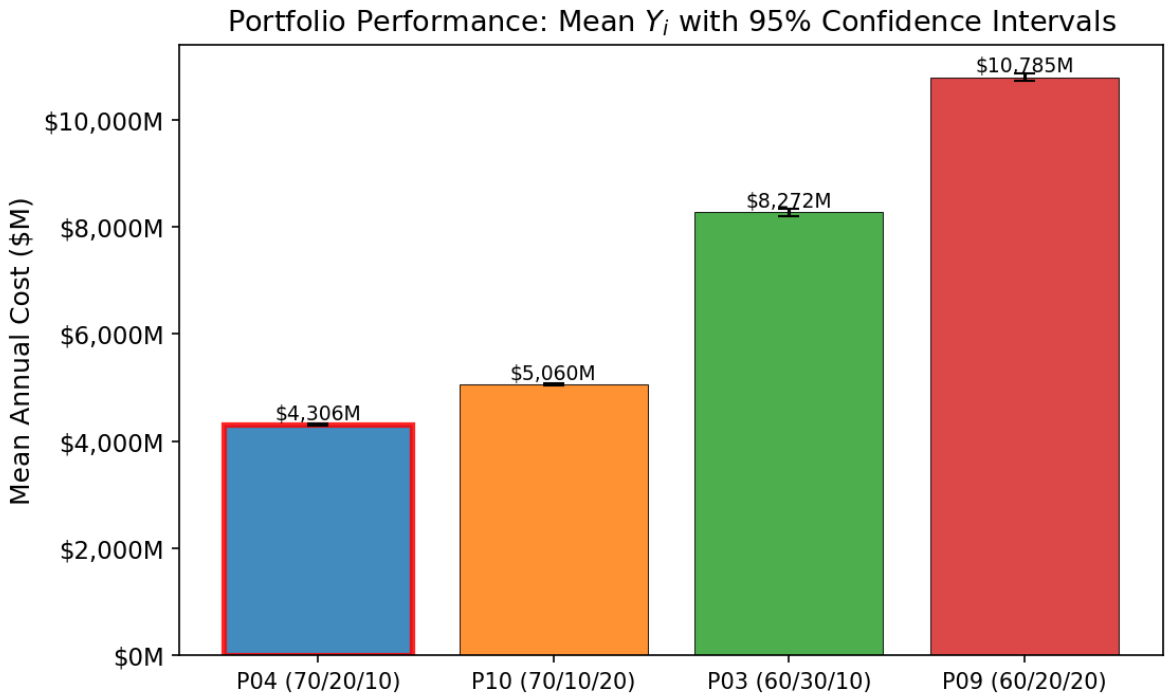


Figure 8: Mean $Y$ with 95% CIs for top-4 portfolios. The firm-fraction gradient drives cost: portfolios with less firm capacity incur large adequacy penalties.

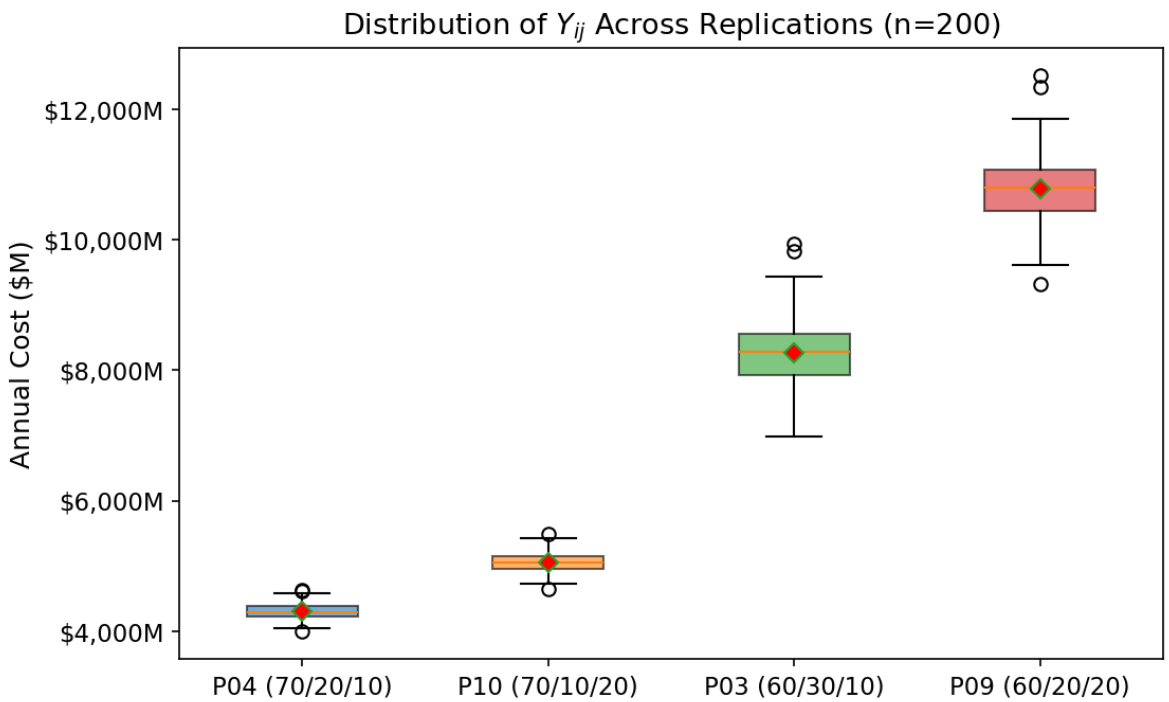


Figure 9: Distribution of $Y_{ij}$ for top-4 portfolios. P03 and P09 exhibit heavy right tails from high-penalty replications during sustained low-renewable periods.

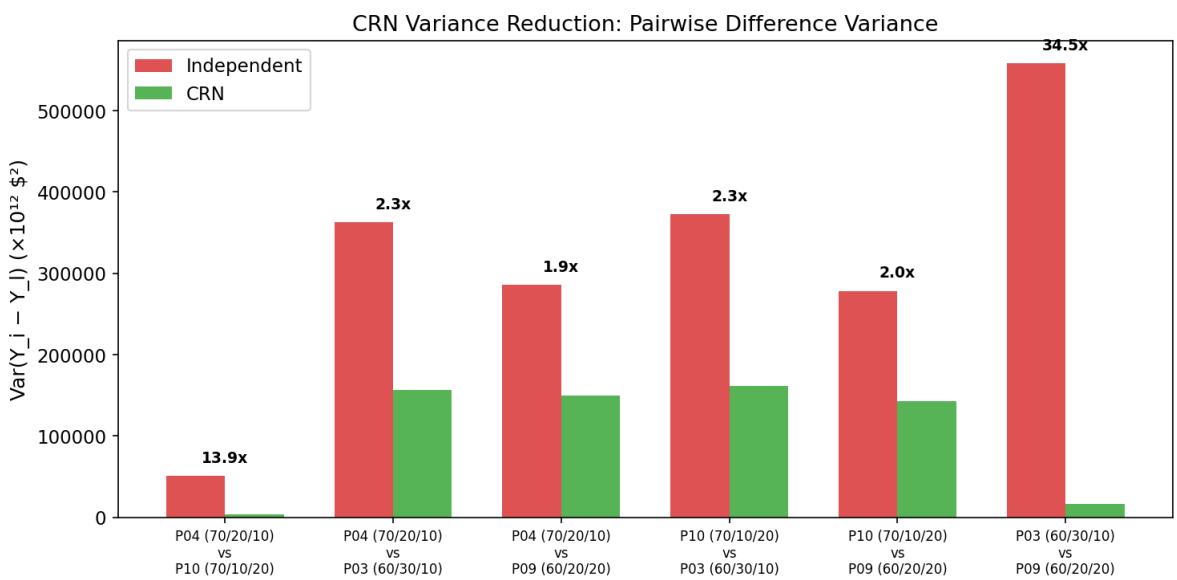


Figure 10: CRN variance reduction: pairwise difference variance under independent vs. CRN sampling. Efficiency ranges from 2× (P04 vs. P09) to 35× (P03 vs. P09).

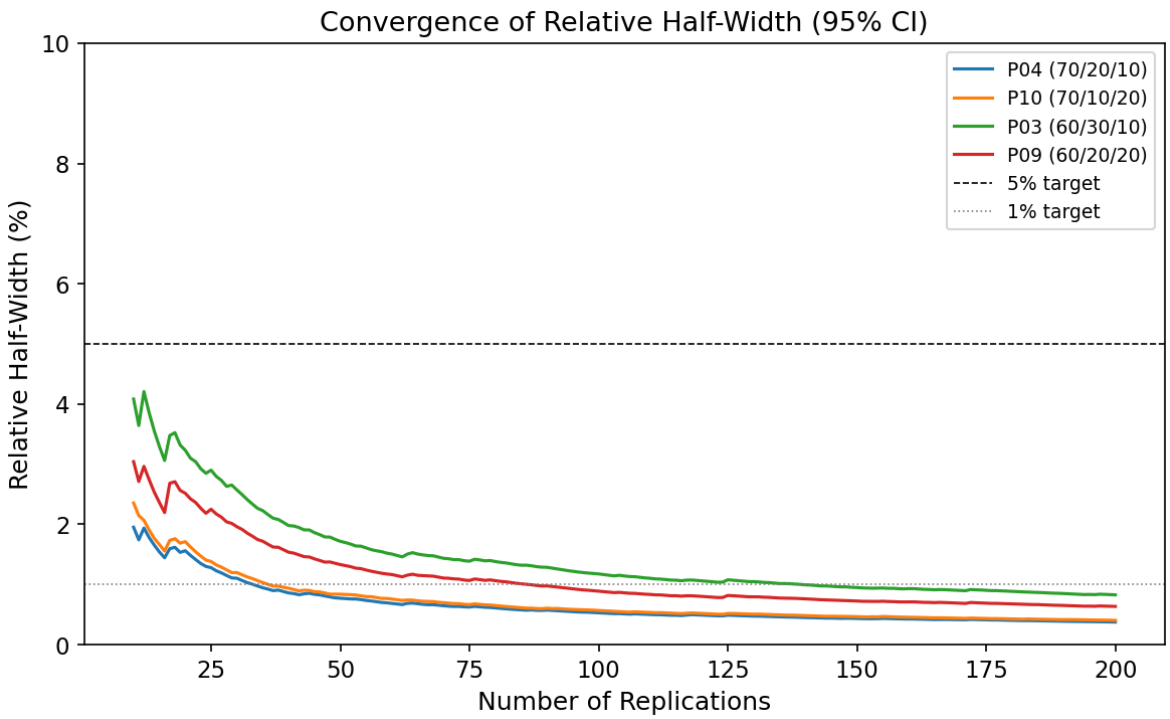


Figure 11: Relative half-width convergence. All top-4 portfolios achieve RHW $< 1\%$ by $n = 200$; the 5% target is met by $n \approx 15$.

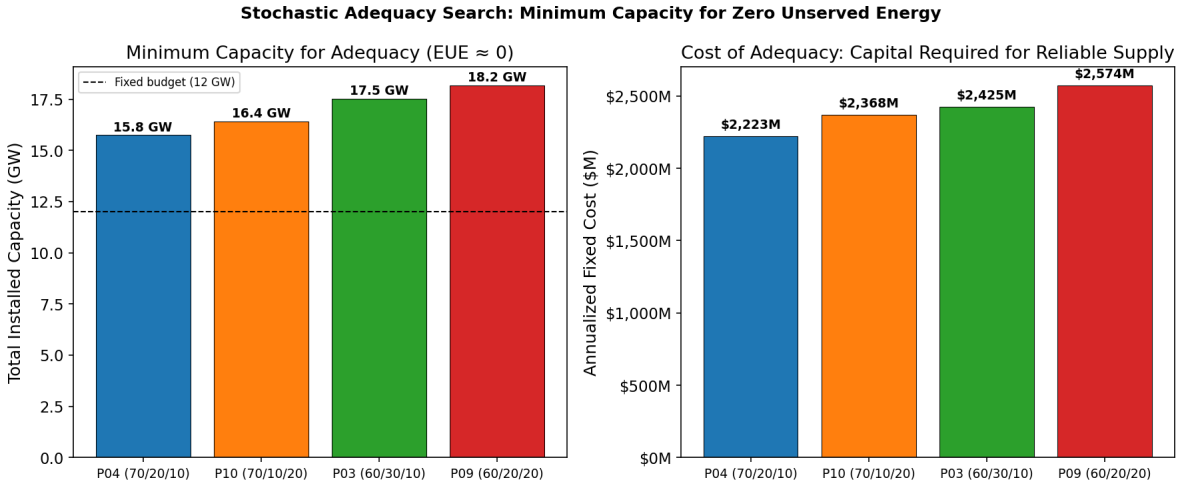


Figure 12: Stochastic adequacy search. Left: minimum capacity for EUE $\approx 0$. Right: annualized fixed cost at adequate capacity. P04 achieves adequacy at the lowest capacity and cost.

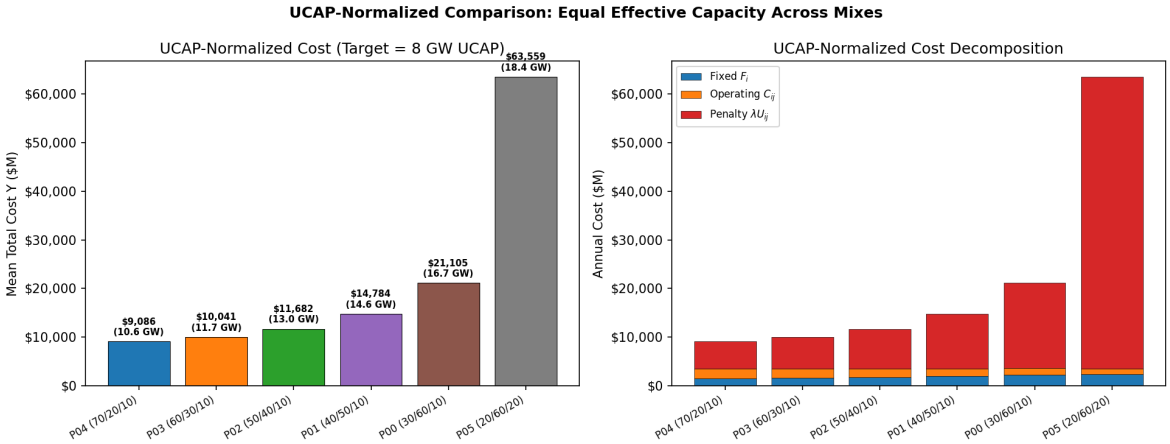


Figure 13: UCAP-normalized comparison. Even at equal effective capacity (8 GW UCAP), P04 remains cheapest. Low-firm portfolios incur large adequacy penalties because ELCC does not capture sustained low-renewable events.

# G Code Architecture

The simulation is implemented in Python 3.11 (NumPy, SciPy). Twelve source files total approximately 2,500 lines. The full pipeline runs in approximately 5 minutes on a consumer laptop. All results are reproducible via `python src/main.py`.

Table 7: Code modules.

| Module | Responsibility |
|---|---|
| `config.py` | Parameters, portfolio generation, ELCC |
| `crn.py` | CRN via SeedSequence, NORTA AR(1) |
| `dispatch.py` | Hourly merit-order dispatch |
| `kn.py` | Kim–Nelson sequential elimination |
| `experiment.py` | DBG, factorial, Bonferroni CIs, VOLL sensitivity |
| `enhancements.py` | CRN efficiency, decomposition, RHW |
| `input_fitting.py` | Parametric fitting to CODERS/IESO |
| `adequacy.py` | Stochastic adequacy search |
| `ucap_comparison.py` | UCAP-normalized comparison |
| `plots.py` | Figure generation |
| `main.py` | Pipeline orchestration (10 stages) |

# H Generative AI Disclosure

Generative AI (Claude, Anthropic) was used for assistance with code development, debugging, statistical methodology review, and drafting support. All modeling decisions, parameter choices, experimental design, interpretation of results, and final writing are the author's own.